

Mutual Information-based Preference Disentangling and Transferring for Non-overlapped Multi-target Cross-domain Recommendations

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ABSTRACT

Building high-quality recommender systems is challenging for new services and small companies, because of their sparse interactions. Cross-domain recommendations (CDRs) alleviate this issue by transferring knowledge from data in external domains. However, most existing CDRs leverage data from only a single external domain and serve only two domains. CDRs serving multiple domains require domain-shared entities (i.e., users and items) to transfer knowledge, which significantly limits their applications due to the hardness and privacy concerns of finding such entities. We therefore focus on a more general scenario, non-overlapped multi-target CDRs (NO-MTCDRs), which require no domain-shared entities and serve multiple domains. Existing methods require domain-shared users to learn user preferences and cannot work on NO-MTCDRs. We hence propose MITrans, a novel mutual information-based (MI-based) preference disentangling and transferring framework to improve recommendations for all domains. MITrans effectively leverages knowledge from multiple domains as well as learning both domain-shared and domain-specific preferences without using domain-shared users. In MITrans, we devise two novel MI constraints to disentangle domain-shared and domain-specific preferences. Moreover, we introduce a module that fuses domain-shared preferences in different domains and combines them with domain-specific preferences to improve recommendations. Our experimental results on two real-world datasets demonstrate the superiority of MITrans

in terms of recommendation quality and application range against state-of-the-art overlapped and non-overlapped CDRs.

CCS CONCEPTS

• Information systems → Recommender systems; • Computing methodologies → Learning latent representations.

KEYWORDS

Cross-domain Recommendations; Preferences Disentangling Learning; Mutual Information

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1 INTRODUCTION

Recently, deep learning-based recommendation models have gained wide attention because of their superiority in predicting preferences accurately [26, 29, 45]. Due to the data-hungry nature of deep learning, however, learning deep models with insufficient training data results in impaired performances [44, 52]. To address this issue, cross-domain recommendations (CDRs) have been studied. CDRs can be categorized into single-target, dual-target, and multi-target cases [18, 22, 44]. Most existing efforts consider the former two cases. They focus on scenarios with only two domains and improve recommendation performances on the target domain or on the two domains. Single-target CDRs learn and transfer knowledge based on the data from an external source domain to facilitate the recommendations in the sparse target domain [30, 53]. Dual-target CDRs design dual-learning frameworks to improve recommendations in the source and target domains simultaneously [23, 24].

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Some recent efforts consider that user preferences in CDRs consist of domain-*shared* and domain-*specific* preferences. Their experimental results demonstrate that transferring the domain-specific preferences leads to a negative transfer issue, i.e., the knowledge transferred from the source domain degrades the recommendation performance in the target domain [25, 27, 28, 40]. Some works mitigate the negative transfer issue by learning and transferring domain-shared preferences [5, 22, 32, 39, 46]. Using domain-shared users, they learn a global representation for each user across all domains by aggregating their interactions in each domain, or they learn local representations for each user in each domain and then align these local representations across different domains. By doing so, the global and aligned local representations can learn and transfer domain-shared preferences. Recent studies introduce individual-level mutual information (MI) and Kullback-Leibler (KL) divergence between user or item representations learned in different domains to extract both domain-specific and domain-shared preferences [6, 9]. Given their explicit measurement of domain differences, such MI and KL-divergence methods have empirically shown their superiority in handling the negative transfer issue and improving recommendation performance.

Unfortunately, the above methods still have important challenges in practical usage. (i) **These works leave a challenge in building recommender systems (RSs) that can serve multiple domains and leveraging knowledge from diverse source domains.** This challenge is prevalent across various industries and applications. For example, in e-commerce platforms such as Amazon and eBay, (1) their RSs need to make recommendations for various categories (domains) of products simultaneously, (2) the number of domains is more than two, and (3) exploiting existing single- and dual-target CDRs is not trivial because, to obtain their best performances, we need to test all pairs of domains, which is impractical. (ii) **They rely on domain-shared users, so they are unavailable in non-overlapped CDRs**, i.e., CDRs with neither domain-shared users nor domain-shared items. The requirement for (many) domain-shared users significantly limits their applications, because we may have technological and legal difficulty in identifying users across domains and, given some domains (of even the same platform, e.g., Amazon), it is not guaranteed that we have a sufficient number of domain-shared users for learning. There exist non-overlapped CDRs [7, 31, 36], and they learn domain-shared preferences through aligning learned user representations, learned item representations, and prediction scores across domains at the distribution level. The existing non-overlapped CDRs assume single-target or dual-target scenarios, and their alignment operation only extracts domain-shared preferences or even learns a preference containing strong bias towards domains with dense and rich interaction data. They hence cannot work on multi-target scenarios and produce impaired recommendations due to ignore domain-specific preferences. Also, some works developed multi-target CDRs [5, 22], but they require domain-*shared* users to transfer individual knowledge.

To overcome the above limitations and challenges, we consider a non-overlapped multi-target CDR, which is a general setting because it does not limit the number of domains and does not require domain-shared users (items). To this end, we propose MITrans, a novel MI-based preference disentangling and transferring framework. Inspired by the CLUB [8] upper bound for MI, we propose

a novel anchor-free variant of CLUB, AFCLUB, to measure the MI between the item representations in different domains. The domain-shared preferences can be learned by minimizing AFCLUB, and, different from CLUB, AFCLUB removes the requirement for domain-shared users and items by estimating the MI for all pairs of items from different domains. Moreover, we develop a novel anchor-free domain-contrastive estimation, AFDCE, to serve as a lower bound for the MI between the item representations in different domains, and we maximize AFDCE to learn domain-shared preferences. By measuring the similarities between all pairs of items from the same and different domains, AFDCE removes the requirement for domain-shared users and items and can serve the non-overlapped scenario. We further develop a multi-domain graph transfer to transfer and fuse the learned domain-shared preferences among multiple domains. After that, we combine the domain-shared preferences with the domain-specific preferences in each domain to improve the recommendations in all domains simultaneously. A summary of our contributions is as follows:

- We consider the scenario of non-overlapped multi-target CDRs, which is a ubiquitous scenario in real-world recommendations but has not been studied yet. To study this scenario, we propose MITrans, a novel MI-based preference disentangling and transferring framework.
- In MITrans, we introduce a novel upper bound (AFCLUB) and a novel lower bound (AFDCE) for MI to explicitly measure the MI between user preferences in different domains. Thanks to these new techniques, MITrans can learn both domain-specific and domain-shared preferences without requiring domain-shared users and items.
- We conduct extensive experiments on real-world datasets to show the superiority of MITrans on multi-target CDRs both with and without domain-shared users. Our source code of MITrans is available at <https://github.com/ZL6298/MITrans>.

2 RELATED WORK

Preference-disentangled CDR. DAL (Domain Adversarial Learning) [13] and RSA (Representation Space Alignment) are two mainstream technologies for learning domain-shared preferences. Conventional DAL-based studies construct a domain discriminator to force the feature extractor generating domain-invariant representations for users. DAREC [43] constrains the domain-invariant quality of user preferences by maximizing the cross-entropy between the domain labels predicted by the domain discriminator and the ground truth labels, and it measures the individual cross-entropy over domain-shared users. C²DSR [4] captures domain-invariant preferences by fusing users' source and target interactions and maximizing ML between single- and cross-domain user preferences. RecSys-DAN [36], DPMCDR [12], and TDAR [42] remove the requirement for domain-shared users. They capture domain-invariant preferences by measuring the cross-entropy between all source and target users or model a consistent latent preference distribution shared among source and target users, while TDAR requires user review data to learn representations for users and items.

RSA-based methods align learned user representations, learned item representations, and predictions across domains to capture domain-shared preferences. For example, COAST [47] and HGCN

[14] propose graph-based approaches to align and combine user representations across domains. To benefit recommendations with both domain-shared and domain-specific preferences, SER [9] and DisenCDR [6] introduced ML and KL-divergence to measure the difference between the user representations learned in different domains. However, the aforementioned methods measure differences or learn representations over domain-shared users, so they cannot work on non-overlapped scenarios. Some works remove this requirement by aligning the representation space in a distribution manner. As an instance, CDR-MMT [21] developed regularization to implicitly punish the difference between the contextual features learned in different domains. ESAM [7] aligns the attribution correlation for the learned user representations between source and target domains. CAFF [31] further improves ESAM by adding another distribution alignment constraint. SRTrans [28] developed a task-oriented knowledge distillation approach to capture domain-shared preferences. These previous non-overlapped CDRs lack explicit mechanisms that extract domain-specific preferences, so extending them to the multi-target scenario is difficult.

Multi-target CDRs. Multi-target CDRs have emerged to improve the recommendation performances for multiple domains simultaneously by combining the knowledge in different domains. For example, GA-MTCDR [21] developed a feature combination layer to combine the interaction graph and item similarity information in each domain. HeroGRAPH [10] aggregates the interaction graph of all domains via domain-shared users and constructs a heterogeneous graph to combine the knowledge from each domain. MPF [41], UniCDR [5], and CAT-ART [22] collect each user’s interaction data in all domains and learn both global and local user representations. Specifically, CAT-ART and MPF combine global and local user representations to capture domain-shared and domain-specific preferences, while UniCDR aligns these two types of representations to learn a robust domain-shared preference and ignores the domain-specific preferences. These multi-target methods transfer individual knowledge across domains via domain-shared users, so they are unavailable in the non-overlapped scenario.

3 PROBLEM STATEMENT

Given a set of domains $\mathcal{D} = \{\mathcal{D}^1, \dots, \mathcal{D}^{N_d}\}$, the recommendation task of multi-target CDRs in this work is the top-K recommendation in all domains, where N_d denotes the number of domains. Let $\mathcal{U}^d$ and $\mathcal{V}^d$ respectively denote sets of users and items in $\mathcal{D}^d$. The interaction set between $\mathcal{U}^d$ and $\mathcal{V}^d$ is denoted as $\mathcal{R}^d = \{(u, v) | u \in \mathcal{U}^d, v \in \mathcal{V}^d\}$. Under the scenario of non-overlapped CDRs, there is no overlap of user and item between any domains. In each domain $\mathcal{D}^d$, the top-K recommendation aims to generate personalized rankings from candidate items $\mathcal{V}_u^d$ and recommends the top-K items for each user u in $\mathcal{U}^d$, where $\mathcal{V}_u^d$ is an item set containing items with no interaction with u .

To achieve this, we introduce a prediction model $f(u, v)$, which, given a user u and an item v , predicts a preference score $y_{u,v} = f(u, v)$. Then, for each user u , the candidate items can be ordered based on their preference scores¹. The main notations are summarised in Table 1.

¹Any search algorithms which allow arbitrary user and item embeddings (e.g., [1–3, 16, 33]) are available.

Table 1: Summary of notations

Symbol	Explanation
$\mathcal{D}^d$	A domain
$\mathbf{U}^d$	User embedding matrix in $\mathcal{D}^d$
$\mathbf{V}^d$	Semantic item embedding matrix in $\mathcal{D}^d$
$\phi^d (\phi^s)$	Dom.-specific (dom.-shared) encoder
$\mathbf{E}^d$	Dom.-specific item emb. matrix in $\mathcal{D}^d$ before LGC
$\mathbf{D}^d$	Dom.-specific item emb. matrix in $\mathcal{D}^d$ after LGC
$\mathbf{H}^d$	Dom.-shared item emb. matrix in $\mathcal{D}^d$ before LGC
$\mathbf{S}^d$	Dom.-shared item emb. matrix in $\mathcal{D}^d$ after LGC
$\mathbf{C}^d$	Cluster embedding matrix in $\mathcal{D}^d$
G_c	Cluster-level cross-domain graph
G_o^d	Cluster-item graph in $\mathcal{D}^d$
$\mathcal{C}^d (\mathcal{Q}^d)$	Vertex set in $G_c (G_o^d)$
$\mathcal{A} (\mathcal{A}^d)$	Adjacency matrix in $G_c (G_o^d)$
$\hat{\mathbf{S}}^d$	Enhanced dom.-shared item emb. matrix in $\mathcal{D}^d$

4 MITRANS

Previous CDRs either lack explicit mechanisms that capture both domain-specific and domain-shared preferences or require domain-shared users to transfer knowledge across domains. We remove these limitations and, for non-overlapped multi-target CDRs, propose MITrans, an MI-based preference disentangling and transferring framework. Its overview is shown in Figure 1. MITrans consists of three modules. The numbers below correspond to the modules in Figure 1.

- (1) The semantic embedding module extracts semantic embeddings for items from their text data. These embeddings are leveraged to (i) capture domain-specific and domain-shared preferences in the second module and (ii) construct cross-domain graphs that transfer knowledge across multiple domains.
- (2) In the disentangled preference learning module, we develop domain-shared and domain-specific encoders, which respectively generate domain-shared and domain-specific embeddings for items by inputting the semantic embeddings extracted in the semantic embedding module. We then propose two novel MI-based constraints to disentangle domain-shared and domain-specific preferences from the two types of item embeddings.
- (3) In the preference fusing module, we adaptively cluster items with their semantic embeddings and construct two types of cross-domain graphs. The adjacency matrices of these graphs are defined as the cosine similarities between semantic embeddings of items and item clusters in different domains. Then, we aggregate the structural information in these graphs to enhance the domain-shared item embeddings in each domain. By doing so, domain-shared preferences in diverse domains can be fused into each domain without requiring domain-shared users and items. After that, we combine these enhanced domain-shared preferences with the domain-specific preferences to further improve recommendation performances.

4.1 Semantic Embedding Module

Previous studies have empirically shown that semantic embeddings of items are effective in transferring knowledge between domains with homogeneous or heterogeneity user behaviors [27, 28, 37, 38].

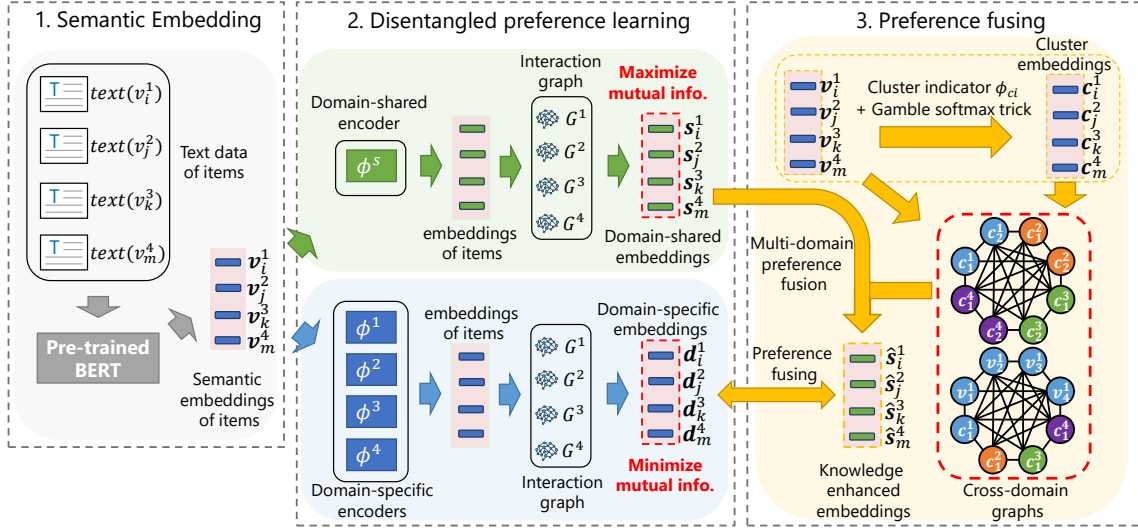


Figure 1: Overview of MITrans

Pre-trained large language models (LLMs) are expected to provide robust token representations containing semantic information [48]. To obtain semantic item embeddings, we employ a prevailing LLM, i.e., BERT[11], and extract token embeddings for each token in the text data of items (item descriptions in this work). Formally, given an item v , we denote its text and semantic embedding as $\text{text}(v)$ and $\mathbf{v}$, respectively, and $\mathbf{v}$ is formulated as:

$$\mathbf{v} = \sum_{t \in \text{text}(v)} \phi_{\text{tf-idf}}(t) \cdot \phi_{\text{BERT}}(t), \quad (1)$$

where $\phi_{\text{tf-idf}}(t)$ is the *tf-idf* score calculated with the corpus containing the text data of all items in a domain. We weight tokens based on their *tf-idf* scores to mitigate the impact of tokens that lack meaning but are frequently present.

To preserve user privacy and increase the applicability of our method, we apply traditional ID embedding for users. For each domain $\mathcal{D}^d$, the embedding for user $u \in \mathcal{U}^d$ is defined as a trainable vector $\mathbf{u}$, which u 's ID can retrieve. Then, the user embeddings in $\mathcal{D}^d$ are denoted as an embedding matrix $\mathbf{U}^d \in \mathbb{R}^{|\mathcal{U}^d| \times N_e}$, where $|\mathcal{U}^d|$ is the number of users in $\mathcal{D}^d$ while N_e is the number of embedding dimensions. Each row in $\mathbf{U}^d$ corresponds to a user embedding.

4.2 Disentangled Preference Learning Module

We propose two MI-based constraints to disentangle domain-shared and domain-specific user preferences. We first introduce a domain-shared encoder and a set of domain-specific encoders to encode semantic item embeddings and generate domain-specific and domain-shared item embeddings. The MI of these two types of embeddings is then measured to optimize these encoders. The optimized encoders are expected to generate embeddings containing domain-shared and domain-specific preferences. Using these preferences has been empirically proven to be superior in alleviating the negative transfer issue and improving recommendations [6, 9, 28]. After that, we employ LGC layers [15] to aggregate multi-hop interaction information into the domain-specific and domain-shared embeddings. The interaction information is obtained from the interaction graph of

each domain. Because no additional parameter is introduced, LGC layers have been proven effective in alleviating the over-fitting issue for sparse domains [28]. Finally, our MI-based constraints are leveraged to disentangle domain-specific and domain-shared preferences based on the two types of item embeddings generated by the two types of encoders.

4.2.1 Domain-specific Preference Learning. We first introduce a domain-specific encoder for each domain to create domain-specific item embeddings from their semantic embeddings. Taking domain $\mathcal{D}^d$ as an example, we denote the domain-specific encoder and the domain-specific item embeddings in this domain as ϕ^d and $\mathbf{E}^d \in \mathbb{R}^{|\mathcal{V}^d| \times N_e}$, respectively. Each row in $\mathbf{E}^d$ is a domain-specific item embedding. For example, the i -th row in $\mathbf{E}^d$ is the domain-specific embedding $\mathbf{e}_i^d$ for item v_i , which is

$$\mathbf{e}_i^d = \phi^d(\mathbf{v}_i), \quad (2)$$

where $\mathbf{v}_i$ is the semantic embedding for item v_i . Then, the user embedding matrix $\mathbf{U}^d$ and the item embedding matrix $\mathbf{E}^d$ are inputted into LGC layers to aggregate multi-hop interaction information with the interaction graph $G^d(\mathbf{U}^d, \mathbf{V}^d; \mathcal{R}^d)$. In this graph, the vertices correspond to all users and items in domain $\mathcal{D}^d$, and the edges are the interactions in $\mathcal{D}^d$. The graph convolutional process on G^d is formulated as

$$(\bar{\mathbf{U}}^d, \mathbf{D}^d) = \text{LGC}(\mathbf{U}^d, \mathbf{E}^d | G^d(\mathbf{U}^d, \mathbf{V}^d; \mathcal{R}^d)), \quad (3)$$

where $\bar{\mathbf{U}}^d$ and $\mathbf{D}^d$ are respectively the domain-specific user and item embeddings after aggregation of the interaction information from G^d .

By minimizing the mutual information of domain-specific item embeddings across domains, we can learn the domain-specific item embeddings containing domain-specific preferences. CLUB [8] is an approximate upper bound of the MI. It is differentiable and hence can be minimized with the gradient descent algorithm. However, CLUB works only on domain-shared users and items because it estimates the MI between two representations of the same object, e.g., two types of embeddings for a user or an item. This is hence

not available in non-overlapped CDRs, so we propose AFCLUB, a novel anchor-free variant of CLUB. AFCLUB measures the mutual information for all item pairs in different domains to remove the identification of domain-shared items, and the AFCLUB between $\mathcal{D}^m$ and $\mathcal{D}^n$ is formulated as

$$\mathcal{L}_{AC}^{m,n} = \frac{1}{N_m N_n} \sum_{i=1}^{N_m} \sum_{j=1, j \neq i}^{N_m} \left(\log p(\mathbf{d}_j | \mathbf{d}_i^m) - \frac{1}{N_n} \sum_{k=1}^{N_n} \log p(\mathbf{d}_k | \mathbf{d}_i^m) \right), \quad (4)$$

where $\mathbf{d}_i^m$ is the i -th row in $\mathbf{D}^m$, whereas N_m and N_n are the numbers of items in $\mathcal{D}^m$ and $\mathcal{D}^n$, respectively. As with [8, 19], we assume that the distribution of the conditional probability $p(\mathbf{d}_j | \mathbf{d}_i)$ follows the normal distribution $\mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\sigma}_i^2)$. The conditional probability $p(\mathbf{d}_j | \mathbf{d}_i)$ represents the probability that the domain-specific embedding after aggregating $\mathbf{G}^d$ for item j is $\mathbf{d}_j$ after observing $\mathbf{d}_i$. When item i and item j are from different domains, minimizing $p(\mathbf{d}_j | \mathbf{d}_i)$ can mitigate the mutual dependence between $\mathbf{d}_i$ and $\mathbf{d}_j$ to extract domain-shared features. We design two domain-shared agent networks ϕ_μ and ϕ_σ to calculate $\boldsymbol{\mu}_i$ and $\log \boldsymbol{\sigma}_i$:

$$\boldsymbol{\mu}_i = \phi_\mu(\mathbf{d}_i), \quad \log \boldsymbol{\sigma}_i = \phi_\sigma(\mathbf{d}_i). \quad (5)$$

Then, $p(\mathbf{d}_j | \mathbf{d}_i)$ can be computed by

$$p(\mathbf{d}_j | \mathbf{d}_i) = -\frac{(\boldsymbol{\mu}_i - \mathbf{d}_j)^2}{2 \cdot \exp(\log \boldsymbol{\sigma}_i)}. \quad (6)$$

4.2.2 Domain-shared Preference Learning. The goal of this part is to learn domain-shared preferences. To achieve this, we first develop a domain-shared encoder ϕ_s to encode semantic item embeddings. Given a domain $\mathcal{D}^d$, we denote domain-shared item embeddings in $\mathcal{D}^d$ as a matrix $\mathbf{H}^d \in \mathbb{R}^{|\mathcal{V}^d| \times N_c}$. Then, the domain-shared embedding for an item $v_i \in \mathcal{V}^d$ is computed as

$$\mathbf{h}_i^d = \phi^s(\mathbf{v}_i), \quad (7)$$

where $\mathbf{v}_i$ is the semantic embedding of item v_i . It is important to notice that the encoder ϕ^s is shared for all domains.

After getting $\mathbf{H}^d$, we input $\mathbf{H}^d$ and the user embedding matrix $\mathbf{U}^d$ into LGC layers to convolute multi-hop interaction information, that is,

$$(\hat{\mathbf{U}}^d, \mathbf{S}^d) = \text{LGC}(\mathbf{U}^d, \mathbf{H}^d | \mathbf{G}^d(\mathbf{U}^d, \mathbf{V}^d; \mathcal{R}^d)), \quad (8)$$

where $\hat{\mathbf{U}}^d$ and $\mathbf{S}^d$ are respectively domain-shared user and item embeddings after aggregating interaction information in $\mathcal{D}^d$.

Then, we learn domain-shared item embeddings containing domain-shared preferences by maximizing the mutual information between these embeddings in different domains. Information noise contrastive estimation (InfoNCE) is a widely used lower bound that estimates the mutual information between two variables [34]. However, the measurement of InfoNCE demands domain-shared users or items as the anchors in CDRs. Therefore, we propose a novel anchor-free domain contrastive estimation (AFDCE) which can measure the mutual information of the domain-shared item embeddings across domains without using domain-shared users. The inverse AFDCE loss between domain $\mathcal{D}^m$ and $\mathcal{D}^n$ is defined as:

$$\mathcal{L}_{iDCE}^{m,n} = -\frac{1}{N_m N_n} \sum_{i=1}^{N_m} \sum_{j=1}^{N_n} \log \frac{\exp(\cos(\mathbf{s}_i^m, \mathbf{s}_j^n) / \tau)}{\frac{1}{N_m} \sum_{k=1, k \neq i}^{N_m} \exp(\cos(\mathbf{s}_i^m, \mathbf{s}_k^n) / \tau)}, \quad (9)$$

where $\cos(\cdot)$ denotes the cosine similarity function and the hyper-parameter τ controls the temperature. By minimizing $\mathcal{L}_{iDCE}^{m,n}$, we can maximize the AFDCE lower bound and also maximize the mutual information between domain-shared item embeddings $\mathbf{S}^m$ and $\mathbf{S}^n$ to capture domain-shared preferences.

4.3 Preference Fusing Module

This module fuses domain-shared preferences in multiple domains and combines the fused domain-shared preferences and the domain-specific preferences to improve recommendation performances. It consists of a multi-domain graph transfer and a task-oriented preference fusion.

4.3.1 Multi-domain Graph Transfer. To transfer and fuse domain-shared preferences across multi-domains, we develop a multi-domain variant of the two-tier graph transfer in SRTrans [28]. We first adaptively cluster the items in each domain by using their semantic embeddings and compute a cluster embedding for each cluster. Following the idea in DE-RRD [17] and SRTrans [28], we apply a cluster indicator net ϕ_{ci} and the widely used Gumbel-Softmax reparameterization trick to achieve the end-to-end cluster. For each domain $\mathcal{D}^d \in \mathcal{D}$, we compute a cluster assignment matrix $\mathbf{P}^d \in \mathbb{R}^{|\mathcal{V}^d| \times N_c}$, where the i -th row of the j -th column in $\mathbf{P}^d$ indicates the probability that item v_i is assigned to the cluster c_j . Note that N_c is the number of clusters. Taking the item v_i as an example, the cluster assignment vector of v_i is formulated by

$$\mathbf{p}_i^d = \phi_{ci}(\mathbf{v}_i). \quad (10)$$

Let $\mathbf{M}^d \in \mathbb{R}^{|\mathcal{V}^d| \times N_c}$ denote the binary gradient matrix for domain $\mathcal{D}^d$. The i -th row in $\mathbf{M}^d$ indicates the cluster assignment result for item v_i and is denoted as $\mathbf{m}_i^d$. By using the Gumbel-Softmax reparameterization trick, $\mathbf{m}_i^d$ can be calculated as

$$\mathbf{m}_i^d = \text{Softmax}\left(\frac{\mathbf{p}_i^d + \mathbf{g}}{\tau}\right), \quad \mathbf{g} \sim \text{Gumbel}(0, 1) \quad (11)$$

where $\mathbf{g} \in \mathbb{R}^{N_c}$ is a Gumbel noise drawn from the Gumbel(0, 1) distribution and τ is a temperature parameter. After getting $\mathbf{M}^d$, we denote the cluster matrix in $\mathcal{D}^d$ as the average semantic embeddings of the items assigned to the corresponding clusters:

$$\mathbf{C}^d = \left(\frac{\mathbf{M}^d}{\sum_k \mathbf{M}_{[:,k]}^d} \right)^T \mathbf{V}^d. \quad (12)$$

After computing the cluster matrix for all domains, we construct a cluster-level cross-domain graph $G_c(\mathbf{C}; \mathcal{A})$ to aggregate the cluster-level domain-shared preferences in diverse domains ($\mathcal{A}$ is shown later). The vertices in G_c contain clusters in all domains, which is $\mathbf{C} = \text{Concat}(\mathbf{C}^1; \dots; \mathbf{C}^{N_d})$. The edges in G_c and their corresponding weights are determined as the cosine similarity with softmax between vertices. For example, the edge weight $\mathcal{A}(\mathbf{c}_i, \mathbf{c}_j)$ between clusters $\mathbf{c}_i$ and $\mathbf{c}_j$ is measured by

$$\mathcal{A}(\mathbf{c}_i, \mathbf{c}_j) = \text{Softmax}(\cos(\mathbf{c}_i, \mathbf{c}_j)). \quad (13)$$

Then, the adjacency matrix in G_c is defined as $\mathcal{A} = \{\mathcal{A}(\mathbf{c}_i, \mathbf{c}_j) \mid \mathbf{c}_i \in \mathbf{C}, \mathbf{c}_j \in \mathbf{C}\}$. (Non-zero weight means that there is an edge.) With the graph $G_c(\mathbf{C}; \mathcal{A})$, we fuse and transfer relational knowledge across

domains by leveraging a graph convolutional network (GCN) [20], which is formulated as

$$\hat{\mathbf{C}} = \text{ReLU}\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{C}\mathbf{W}_c\right). \quad (14)$$

In this formula, $\tilde{\mathbf{A}}$ is the adjacency matrix after adding self-loops. Also, $\mathbf{W}_c$ is the trainable weight matrix in this GCN, and $\tilde{\mathbf{D}}$ is the degree matrix with elements $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Because the cluster embeddings $\mathbf{C}$ are calculated with domain-shared item embeddings, the GCN is expected to aggregate domain-shared preferences in different domains into the enhanced cluster embeddings $\hat{\mathbf{C}}$.

After aggregating the domain-shared preferences into $\hat{\mathbf{C}}$, we further construct a cluster-item graph for each domain $\mathcal{D}^d \in \mathcal{D}$ to aggregate the domain-shared preferences into the domain-shared item embeddings in that domain. Taking domain $\mathcal{D}^d$ as an instance, we denote the cluster-item graph for $\mathcal{D}^d$ as $\mathbf{G}_v^d(\mathbf{Q}^d; \mathcal{B}^d)$. Its vertices consist of the clusters in all domains and the items in $\mathcal{D}^d$, i.e., $\mathbf{Q}^d = \text{Concat}(\hat{\mathbf{C}}; \mathbf{S}^d)$. Its adjacency matrix is defined as the cosine similarities with softmax between vertices, which is $\mathcal{B}^d = \{\mathcal{B}^d(q_i, q_j) \mid q_i, q_j \in \mathbf{Q}^d\}$. With this graph, we next use a GCN to transfer the aggregated knowledge from the cluster embeddings $\hat{\mathbf{C}}$ to the domain-shared item embeddings $\mathbf{S}^d$, which is formulated as

$$\hat{\mathbf{Q}}^d = \text{ReLU}\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathcal{B}}^d\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{Q}^d\mathbf{W}_v^d\right), \quad (15)$$

where $\tilde{\mathcal{B}}^d$ is the adjacency matrix with self-loops and $\tilde{\mathbf{D}}$ is the degree matrix on $\tilde{\mathcal{B}}^d$. Also, $\mathbf{W}_v^d$ is the trainable weight matrix in this GCN. Then, we obtain the enhanced domain-shared item embeddings $\hat{\mathbf{S}}^d$, containing aggregated domain-shared preferences from multiple domains, as the last $|\mathcal{V}^d|$ rows of $\hat{\mathbf{Q}}^d$.

4.3.2 Task-oriented Preference Fusion. We achieve preference fusion by distilling domain-shared preferences from the enhanced item embeddings $\hat{\mathbf{S}}^d$ into the domain-specific item embeddings $\mathbf{D}^d$ for each domain $\mathcal{D}^d \in \mathcal{D}$. We compute two types of preference scores with the domain-specific user embeddings $\tilde{\mathbf{U}}^d$ and the two types of item embeddings $\hat{\mathbf{S}}^d$ and $\mathbf{D}^d$. For simplicity, we explain the computation with the example of an interaction (u_i, v_j) . Formally, the two types of preference scores are given by the inner product between the user embedding and the corresponding item embedding:

$$\hat{y}_{i,j}^d = f(\tilde{\mathbf{u}}_i^d, \mathbf{d}_j^d) \quad \text{and} \quad \tilde{y}_{i,j}^d = f(\tilde{\mathbf{u}}_i^d, \hat{\mathbf{s}}_j^d). \quad (16)$$

Then, the preference fusion loss is defined as the KL-divergence between the two types of preference scores, which is

$$\mathcal{L}_{PF}^d = \sum_{(u_i, v_j) \in \mathcal{R}^d} \left(\hat{y}_{i,j}^d \log \frac{\hat{y}_{i,j}^d}{\tilde{y}_{i,j}^d} + (1 - \hat{y}_{i,j}^d) \log \frac{(1 - \hat{y}_{i,j}^d)}{(1 - \tilde{y}_{i,j}^d)} \right). \quad (17)$$

After minimizing $\mathcal{L}_{PF}^d$, the domain-shared preferences and the domain-specific preferences can be fused into the domain-specific item embeddings.

4.4 Learning Objective

We employ the BPR [35] loss to measure the performance of predictions. For our multi-target CDR, the prediction objective is to simultaneously minimize the BPR loss in all domains.

For each domain $\mathcal{D}^d \in \mathcal{D}$, we extend its interaction set to $\mathcal{R}^d = \{(u_i, v_j, v_k) \mid u_i \in \mathcal{U}^d, v_j \in \mathcal{V}^d, v_k \in \mathcal{V}_u^d\}$ by randomly sampling a negative item v_k for each interaction (u_i, v_j) . Then, the BPR loss of $\mathcal{D}^d$ is measured by

$$\mathcal{L}_{BPR}^d = - \sum_{(u_i, v_j, v_k) \in \mathcal{R}^d} \ln \sigma(\hat{y}_{i,j}^d - \hat{y}_{i,k}^d), \quad (18)$$

where $\ln \sigma$ is the logistic sigmoid function. Finally, the total loss of MITrans is defined as

$$\mathcal{L} = \sum_{m=1}^{N_d} \left(\mathcal{L}_{BPR}^m + \alpha \mathcal{L}_{PF}^m + \sum_{n=1, n \neq m}^{N_d} \left(\beta \mathcal{L}_{AC}^{m,n} + \gamma \mathcal{L}_{DCE}^{m,n} \right) \right), \quad (19)$$

where α , β , and γ are hyper-parameters that control the weights of each component. In $\mathcal{L}$, minimizing the inverse AFDCE loss $\mathcal{L}_{DCE}^{m,n}$ and the AFCLUB loss $\mathcal{L}_{AC}^{m,n}$ is expected to disentangle the domain-shared and domain-specific preferences. Then, these preferences are combined into domain-specific item embeddings by minimizing the preference fusion loss $\mathcal{L}_{PF}^m$.

5 EXPERIMENT

5.1 Datasets

Amazon dataset² and Douban dataset³ are two widely used benchmarks for CDR evaluation [49–51]. Considering the non-overlapped multi-target CDR task in this work, we selected 4 categories of rating data from Amazon dataset, including Books, Movies and TV (Movies), Sports and Outdoors (Sports), and Video Games (Games), as different domains. We also selected 3 categories of rating data from Douban dataset, including Books, Movies, and Musics, as different domains. Table 2 reports the statistical information of these datasets.

5.1.1 Amazon. For each domain in Amazon, we collected the five-score ratings on products from 30/6/2017 to 3/10/2018 and descriptions of products in the same period as interaction set and text data of items. In the data pre-processing stage, we first filtered out the interactions for items with no text data. Then, we constructed two CDR scenarios on Amazon dataset, including a sparse scenario and a relatively dense scenario, to evaluate the performance of MITrans in both sparse and dense settings. For the sparse scenario, we sampled 8,000 users with 3 to 5 ratings in each domain to adapt to a sparse setting. On the other hand, for the dense scenario, we sampled 8,000 users with 5 to 15 ratings in each domain to form a relatively dense environment. We also built an overlap scenario on Amazon dataset by identifying domain-shared users in 3 domains, i.e., Books, Movies, and Sports, to compare with CDRs requiring domain-shared users. It is important to note that there is no domain-shared user overlapping in all the 4 domains.

5.1.2 Douban. For each domain in Douban, we collected all ratings with the score larger than 3 and tags of products as interaction set and text data of items. In the data pre-processing stage, we first filtered out the interactions for items with no text data. We sampled items with 5 to 30 ratings in each domain to form a dense environment.

²jmcauley.ucsd.edu/data/amazon/

³kaggle.com/datasets/fengzhujoe/douban-datasetratingreviewside-information/

Table 2: Basic information on datasets we used. Int./U (Int./I) is the average number of interactions per user (item).

Scenario	Domain	Users	Items	Int.	Int./U	Int./I
Amazon						
Sparse	Books	8,000	9,511	39,861	5.0	4.2
	Movies	8,000	6,468	34,448	4.3	5.3
	Sports	8,000	6,831	32,000	4.0	4.7
	Games	8,000	4,269	24,689	3.1	5.8
Dense	Books	8,000	11,900	6,3188	7.9	5.3
	Movies	8,000	8,490	55,735	7.0	6.6
	Sports	8,000	11,749	43,999	5.5	3.7
	Games	6,650	6,659	45,157	6.8	6.8
Overlap	Books		10,817	25,432	9.6	2.4
	Movies	2,646	6,050	22,988	8.7	3.8
	Sports		6,623	16,208	6.1	2.4
Douban						
Dense	Books	940	5,463	29,921	31.8	5.5
	Movies	1,342	9,615	74,410	55.4	7.7
	Musics	682	4,925	27,199	39.9	5.5

5.2 Evaluated Method

We compared MITrans with the following state-of-the-art methods, including single-domain, non-overlapped, and overlapped CDRs.

5.2.1 Single-domain Methods. We first made comparisons with LightGCN [15], a state-of-the-art graph-based method. LightGCN designs a novel light graph convolutional layer to extract graph knowledge and alleviate the over-fitting issue effectively.

5.2.2 Non-overlapped CDR Methods. In this branch of CDR methods, we compared with RecSys-DAN [36], ESAM [7], and CFAA [31]. These methods capture domain-shared preferences by aligning the distribution of user representations, item representations, or predictions between domains. We also evaluated a recent graph-based CDR method, i.e., SRTrans [28], which learns and transfers relational knowledge between domains to alleviate the negative transfer issue.

5.2.3 Overlapped CDR Methods. COAST [47] and HGCN [14] are state-of-the-art graph-based efforts that learn domain-shared preferences by aligning or fusing user and item preferences between domains. They also construct cross-domain heterogeneous (hyper-)graphs to aggregate high-order graph knowledge. We also compared MITrans with UniCDR [5] and CAT-ART [22], which are state-of-the-art multi-target CDR methods. UniCDR and CAT-ART learn global and local representations for users. The former captures domain-shared preferences by aligning these two types of representations, whereas the latter combines these two representations to capture both domain-shared and domain-specific preferences.

5.3 Evaluation Criteria

We measured performances of recommendations for all domains. For each user u in each domain $\mathcal{D}^d$, we took their last and second-last interactions to serve as the test and the validation data, respectively. The remaining interactions were used as the training data. Then, we randomly selected 99 negative items from $\bar{\mathcal{V}}_u^d$ and ranked

them with the test item based on the calculated preference scores. After that, we recommended the top-K items in the ranking to this user. The performance of the top-K recommendations was measured by the widely used *Hit Ratio* (HR) and *Normalized Discounted Cumulative Gain* (NDCG).

For each domain, we experimented 10 groups of negative items and reported the average performance with the 95% confidence level. We ran experiments for every pair of domains for all single-target and dual-target CDR methods (RecSys-DAN, ESAM, CFAA, SRTrans, COAST, and HGCN). The result on a given dataset is the best performance among all combinations.

5.4 Implementation Details

The codes of LightGCN⁴, UniCDR⁵, and CAT-ART⁶ were obtained from the corresponding GitHub repositories. For fair comparisons, we replaced the item embeddings for all evaluated methods with our semantic item embedding that can be calculated by Equation 1. We also unified the base model for all non-overlapped methods as the same LightGCN model. We used the same mini-batch mechanism and Adam optimizer with a learning rate of 0.01 to train all evaluated methods, where the batch size was 2048. We also set the embedding size N_e to 32, 64, and 32 for all evaluated methods in the corresponding sparse, dense, and overlapping scenarios. For all non-overlapped methods, the number of LGC layers in the base model was set to 2, 3, and 1 for the sparse, dense, and overlapping scenarios, respectively. The setting of the remaining hyper-parameters in the baselines followed their original papers.

In MITrans, we built the two encoders ϕ^d and ϕ^s , the two agent networks ϕ_μ and ϕ_σ , and the cluster indicator net ϕ_{ci} as a dense layer with the shape of $[N_e, N_e]$. The temperature factor τ and the weight of the preference fusion loss α were 0.0001 and 0.1, respectively. The weight of AFCLUB loss β was 0.01, 0.1, and 0 for the experiments in the sparse, dense, and overlapping scenarios, respectively. The weight of AFDCE loss was respectively 0, 0.1, and 0.01 for the corresponding sparse, dense, and overlapping scenarios. These hyper-parameters were set based on the average performance of all domains over the validation set.

5.5 Experiment Result

5.5.1 Performance Comparisons. Table 3 and Table 4 report the recommendation performances of evaluated methods on Amazon dataset and Douban dataset, respectively. From these tables, we have the following observations.

(1) MITrans outperforms all non-overlapped methods in all scenarios of Amazon dataset and achieves the best and the second-best performances in the dense scenario of Douban dataset. For example, in the sparse scenario of Amazon, MITrans gains an average of 16.5% improvement on HR@5 and an average of 19.6% improvement on NDCG@5. In the dense scenario of Amazon, the average improvement of MITrans is 8.5% on HR@5 and 5.3% on NDCG@5. These improvements are given by the multi-domain graph transfer module and the two MI-based constraints, i.e., AFCLUB and AFDCE. The results demonstrate their effectiveness in transferring

⁴<https://github.com/kuandeng/LightGCN>

⁵<https://github.com/cjx96/UniCDR>

⁶<https://github.com/Chain123/CAT-ART>

Table 3: Comparison result on Amazon. Performances (%) $\pm$ 95% confidence intervals (%) are reported. Bold shows the winner, and $\downarrow$ indicates the result suffering negative transfer compared with LightGCN.

Amazon		Books		Movies		Sports		Games	
Scenario	Method	HR@5	NDCG@5	HR@5	NDCG@5	HR@5	NDCG@5	HR@5	NDCG@5
Sparse	LightGCN	19.8 $\pm$ 0.2	13.4 $\pm$ 0.2	14.8 $\pm$ 0.2	9.9 $\pm$ 0.1	9.0 $\pm$ 0.1	6.1 $\pm$ 0.1	10.3 $\pm$ 0.2	6.6 $\pm$ 0.1
	RecSys-DAN	8.7 $\pm$ 0.1 $\downarrow$	5.7 $\pm$ 0.1 $\downarrow$	9.0 $\pm$ 0.1 $\downarrow$	5.4 $\pm$ 0.1 $\downarrow$	12.2 $\pm$ 0.2	9.1 $\pm$ 0.1	9.1 $\pm$ 0.1 $\downarrow$	5.8 $\pm$ 0.1 $\downarrow$
	ESAM	15.8 $\pm$ 0.2 $\downarrow$	10.9 $\pm$ 0.1 $\downarrow$	12.6 $\pm$ 0.2 $\downarrow$	8.3 $\pm$ 0.1 $\downarrow$	8.4 $\pm$ 0.1 $\downarrow$	5.6 $\pm$ 0.1 $\downarrow$	10.2 $\pm$ 0.2 $\downarrow$	6.4 $\pm$ 0.1 $\downarrow$
	CFAA	11.8 $\pm$ 0.2 $\downarrow$	7.4 $\pm$ 0.1 $\downarrow$	11.4 $\pm$ 0.2 $\downarrow$	7.1 $\pm$ 0.1 $\downarrow$	16.3 $\pm$ 0.1	11.4 $\pm$ 0.1	11.0 $\pm$ 0.2	6.9 $\pm$ 0.1
	SRTrans	19.1 $\pm$ 0.2 $\downarrow$	13.9 $\pm$ 0.1	14.5 $\pm$ 0.1 $\downarrow$	10.1 $\pm$ 0.1	9.2 $\pm$ 0.1	6.4 $\pm$ 0.1	10.4 $\pm$ 0.1	6.8 $\pm$ 0.1
	MITrans	21.6 $\pm$ 0.1	15.3 $\pm$ 0.1	15.2 $\pm$ 0.1	10.4 $\pm$ 0.1	20.7 $\pm$ 0.1	14.4 $\pm$ 0.1	14.0 $\pm$ 0.1	9.4 $\pm$ 0.1
	w/o SE	18.2 $\pm$ 0.1 $\downarrow$	13.3 $\pm$ 0.1 $\downarrow$	13.2 $\pm$ 0.1 $\downarrow$	9.3 $\pm$ 0.1 $\downarrow$	7.1 $\pm$ 0.1 $\downarrow$	5.0 $\pm$ 0.1 $\downarrow$	7.2 $\pm$ 0.1 $\downarrow$	4.4 $\pm$ 0.1 $\downarrow$
	w/o DPL	21.3 $\pm$ 0.1	15.1 $\pm$ 0.1	15.1 $\pm$ 0.1	10.3 $\pm$ 0.1	16.5 $\pm$ 0.2	12.0 $\pm$ 0.1	13.4 $\pm$ 0.2	9.1 $\pm$ 0.1
	w/o PF	21.7 $\pm$ 0.1	15.3 $\pm$ 0.1	15.3 $\pm$ 0.1	10.5 $\pm$ 0.1	21.2 $\pm$ 0.2	14.9 $\pm$ 0.1	13.5 $\pm$ 0.1	9.3 $\pm$ 0.1
Dense	LightGCN	24.6 $\pm$ 0.2	16.3 $\pm$ 0.1	17.1 $\pm$ 0.2	11.1 $\pm$ 0.2	13.5 $\pm$ 0.1	9.5 $\pm$ 0.1	18.6 $\pm$ 0.2	11.9 $\pm$ 0.2
	RecSys-DAN	8.1 $\pm$ 0.2 $\downarrow$	4.8 $\pm$ 0.1 $\downarrow$	7.4 $\pm$ 0.1 $\downarrow$	4.6 $\pm$ 0.1 $\downarrow$	7.8 $\pm$ 0.1 $\downarrow$	5.2 $\pm$ 0.1 $\downarrow$	8.9 $\pm$ 0.1 $\downarrow$	5.5 $\pm$ 0.1 $\downarrow$
	ESAM	14.2 $\pm$ 0.2 $\downarrow$	8.9 $\pm$ 0.1 $\downarrow$	10.1 $\pm$ 0.2 $\downarrow$	6.3 $\pm$ 0.1 $\downarrow$	9.0 $\pm$ 0.1 $\downarrow$	6.1 $\pm$ 0.1 $\downarrow$	11.8 $\pm$ 0.2 $\downarrow$	7.2 $\pm$ 0.1 $\downarrow$
	CFAA	26.0 $\pm$ 0.2	18.6 $\pm$ 0.1	17.4 $\pm$ 0.1	11.4 $\pm$ 0.1	11.6 $\pm$ 0.2 $\downarrow$	7.6 $\pm$ 0.1 $\downarrow$	10.6 $\pm$ 0.2 $\downarrow$	6.5 $\pm$ 0.1 $\downarrow$
	SRTrans	27.3 $\pm$ 0.2	19.4 $\pm$ 0.2	18.1 $\pm$ 0.2	12.2 $\pm$ 0.1	13.7 $\pm$ 0.1	10.0 $\pm$ 0.1	20.8 $\pm$ 0.2	14.0 $\pm$ 0.2
	MITrans	28.7 $\pm$ 0.1	20.3 $\pm$ 0.1	18.3 $\pm$ 0.1	12.3 $\pm$ 0.0	16.4 $\pm$ 0.2	10.8 $\pm$ 0.1	22.5 $\pm$ 0.1	15.1 $\pm$ 0.1
	w/o SE	24.8 $\pm$ 0.1	18.4 $\pm$ 0.1	16.0 $\pm$ 0.1 $\downarrow$	11.2 $\pm$ 0.1	11.2 $\pm$ 0.1 $\downarrow$	8.5 $\pm$ 0.1 $\downarrow$	17.1 $\pm$ 0.1 $\downarrow$	11.8 $\pm$ 0.1 $\downarrow$
	w/o DPL	26.8 $\pm$ 0.1	19.1 $\pm$ 0.1	17.8 $\pm$ 0.1	11.9 $\pm$ 0.1	15.5 $\pm$ 0.1	11.1 $\pm$ 0.1	21.0 $\pm$ 0.2	13.9 $\pm$ 0.1
	w/o PF	19.0 $\pm$ 0.2 $\downarrow$	12.4 $\pm$ 0.1 $\downarrow$	15.3 $\pm$ 0.1 $\downarrow$	9.8 $\pm$ 0.1 $\downarrow$	11.1 $\pm$ 0.1 $\downarrow$	7.5 $\pm$ 0.1 $\downarrow$	17.4 $\pm$ 0.1 $\downarrow$	11.3 $\pm$ 0.1 $\downarrow$
Overlap	LightGCN	10.5 $\pm$ 0.2	7.0 $\pm$ 0.1	11.1 $\pm$ 0.2	6.9 $\pm$ 0.1	8.4 $\pm$ 0.2	5.9 $\pm$ 0.1		
	CFAA	9.0 $\pm$ 0.3 $\downarrow$	5.7 $\pm$ 0.1 $\downarrow$	8.3 $\pm$ 0.2 $\downarrow$	5.3 $\pm$ 0.1 $\downarrow$	9.8 $\pm$ 0.2	6.6 $\pm$ 0.1		
	SRTrans	10.2 $\pm$ 0.2 $\downarrow$	6.6 $\pm$ 0.1 $\downarrow$	9.9 $\pm$ 0.2 $\downarrow$	6.5 $\pm$ 0.1 $\downarrow$	9.9 $\pm$ 0.1	6.9 $\pm$ 0.1		
	COAST	14.3 $\pm$ 0.4	8.7 $\pm$ 0.2	10.0 $\pm$ 0.2 $\downarrow$	6.1 $\pm$ 0.1 $\downarrow$	9.3 $\pm$ 0.2	5.5 $\pm$ 0.1 $\downarrow$		
	HGCN	11.8 $\pm$ 0.2	8.1 $\pm$ 0.1	9.8 $\pm$ 0.2 $\downarrow$	6.4 $\pm$ 0.1 $\downarrow$	11.8 $\pm$ 0.1	8.7 $\pm$ 0.1		
	UniCDR	10.7 $\pm$ 0.4	6.2 $\pm$ 0.2 $\downarrow$	9.0 $\pm$ 0.2 $\downarrow$	5.7 $\pm$ 0.2 $\downarrow$	8.6 $\pm$ 0.3	5.2 $\pm$ 0.2 $\downarrow$		
	CAT-ART	16.0 $\pm$ 0.2	10.5 $\pm$ 0.2	10.3 $\pm$ 0.2 $\downarrow$	6.5 $\pm$ 0.1 $\downarrow$	13.1 $\pm$ 0.2	8.5 $\pm$ 0.1		
	MITrans	12.9 $\pm$ 0.2	8.9 $\pm$ 0.1	12.5 $\pm$ 0.2	8.5 $\pm$ 0.1	11.4 $\pm$ 0.2	8.0 $\pm$ 0.2		
	w/o SE	6.6 $\pm$ 0.1 $\downarrow$	4.9 $\pm$ 0.1 $\downarrow$	9.6 $\pm$ 0.1 $\downarrow$	6.4 $\pm$ 0.1 $\downarrow$	6.0 $\pm$ 0.1 $\downarrow$	4.6 $\pm$ 0.1 $\downarrow$		
	w/o DPL	12.6 $\pm$ 0.1	8.7 $\pm$ 0.1	12.3 $\pm$ 0.2	8.3 $\pm$ 0.1	11.3 $\pm$ 0.2	8.0 $\pm$ 0.1		
	w/o PF	12.6 $\pm$ 0.2	8.8 $\pm$ 0.1	12.2 $\pm$ 0.2	8.3 $\pm$ 0.1	12.3 $\pm$ 0.2	8.7 $\pm$ 0.1		

Table 4: Comparison result on Douban. Performances (%) $\pm$ 95% confidence intervals (%) are reported. Bold shows the winner, and $\downarrow$ indicates the result suffering negative transfer compared with LightGCN.

Douban		Books		Movies		Musics	
Scenario	Method	HR@5	NDCG@5	HR@5	NDCG@5	HR@5	NDCG@5
Dense	LightGCN	17.5 $\pm$ 0.5	11.0 $\pm$ 0.3	19.3 $\pm$ 0.2	11.9 $\pm$ 0.1	15.7 $\pm$ 0.3	10.4 $\pm$ 0.2
	RecSys-DAN	5.4 $\pm$ 0.3 $\downarrow$	3.4 $\pm$ 0.2 $\downarrow$	4.0 $\pm$ 0.2 $\downarrow$	2.1 $\pm$ 0.1 $\downarrow$	5.5 $\pm$ 0.4 $\downarrow$	3.3 $\pm$ 0.3 $\downarrow$
	ESAM	13.2 $\pm$ 0.4 $\downarrow$	7.9 $\pm$ 0.2 $\downarrow$	17.4 $\pm$ 0.3 $\downarrow$	10.7 $\pm$ 0.2 $\downarrow$	10.4 $\pm$ 0.4 $\downarrow$	7.1 $\pm$ 0.3 $\downarrow$
	CFAA	18.3 $\pm$ 0.6	11.3 $\pm$ 0.3	17.8 $\pm$ 0.5	10.6 $\pm$ 0.3	15.1 $\pm$ 0.3 $\downarrow$	8.8 $\pm$ 0.2 $\downarrow$
	SRTrans	19.7 $\pm$ 0.4	12.5 $\pm$ 0.2	16.9 $\pm$ 0.4 $\downarrow$	10.8 $\pm$ 0.2 $\downarrow$	18.4 $\pm$ 0.4	11.9 $\pm$ 0.2
	MITrans	20.9 $\pm$ 0.3	13.6 $\pm$ 0.2	20.2 $\pm$ 0.3	12.6 $\pm$ 0.3	18.3 $\pm$ 0.6	11.8 $\pm$ 0.3
	w/o SE	6.9 $\pm$ 0.2 $\downarrow$	4.2 $\pm$ 0.1 $\downarrow$	0.9 $\pm$ 0.3 $\downarrow$	5.8 $\pm$ 0.2 $\downarrow$	7.1 $\pm$ 0.5 $\downarrow$	4.1 $\pm$ 0.2 $\downarrow$
	w/o DPL	17.0 $\pm$ 0.5 $\downarrow$	10.8 $\pm$ 0.2 $\downarrow$	10.9 $\pm$ 0.2 $\downarrow$	6.5 $\pm$ 0.1 $\downarrow$	14.5 $\downarrow$ $\pm$ 0.3	9.2 $\pm$ 0.2 $\downarrow$
	w/o PF	18.5 $\pm$ 0.4	12.1 $\pm$ 0.3	18.7 $\pm$ 0.4 $\downarrow$	11.8 $\pm$ 0.2 $\downarrow$	17.8 $\pm$ 0.5	11.3 $\pm$ 0.3

knowledge among multiple domains and learning domain-specific and domain-shared preferences.

(2) Almost all non-overlapped baselines, excepting SRTrans, suffer the negative transfer issue, particularly in the sparse scenario of Amazon dataset. This is because these methods directly align the

distribution of user or item representations in different domains and ignore domain-specific preferences. As a result, these methods learn only the domain-shared preferences or even learn the preferences with a strong domain bias, particularly when data are sparse. MITrans and SRTrans learn domain-shared and domain-specific

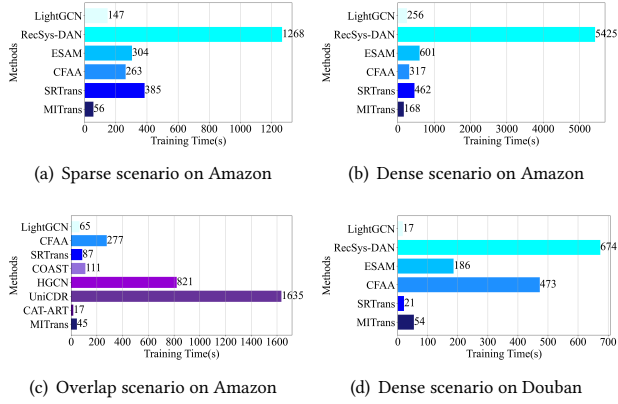


Figure 2: Comparison result of training time (seconds). Average of training time over all domains are reported.

preferences and transfer only the domain-shared preferences to alleviate the negative transfer issue. In particular, MITrans explicitly disentangles user preferences through AFCLUB and AFDCE constraints and has a strong ability to transfer knowledge among multiple domains. These components make MITrans perform best in mitigating the negative transfer issue.

(3) In the overlap scenario, MITrans outperforms most overlapping CDRs, including COAST, HGCN, and UniCDR, in most cases. This result demonstrates that MITrans can achieve competitive with or even better performances than CDRs requiring domain-shared information. Although CAT-ART outperforms MITrans on Books and Sports, its application is significantly limited by the requirement for domain-shared users.

5.5.2 Ablation Study. To study the contribution of each module in MITrans, we compared MITrans with its variants, including (1) w/o SE, which removes the semantic embedding module and uses the ID embedding for items, (2) w/o DPL, which removes the AFCLUB and the AFDCE constraints (i.e., without $\mathcal{L}_{AC}$ and $\mathcal{L}_{iDCE}$), and (3) w/o PF, which cuts out the preference fusing module from MITrans.

The results on Amazon dataset and Douban dataset are reported respectively in Table 3 and Table 4. From these tables, we find the following observations. (1) All modules contribute to the final preference in most cases. (2) MITrans w/o SE performs the worst. This result shows that semantic feature is essential in improving performance in CDRs, particularly when transferring knowledge based on items. (3) MITrans w/o PF performs better than MITrans in the sparse scenario of Amazon dataset. This variant ignores the domain-shared preferences but improves its performance. This means that the domain-shared preferences degrade the final performance, and this result also explains why non-overlapped baselines, which extract only domain-shared preferences, suffer from the negative transfer issue in the same scenario.

5.5.3 Training Efficiency. In this part, we discuss the training time for all comparisons, where the training time indicates the time of training a model until the loss converges on the validation set. All codes run on a Windows PC with a 11th Gen Intel(R) Core(TM) i9-11900KF CPU and a NVIDIA GeForce RTX 3090 GPU.

Figure 2 illustrates the comparison result of training time on Amazon and Douban datasets. The result shows that MITrans takes

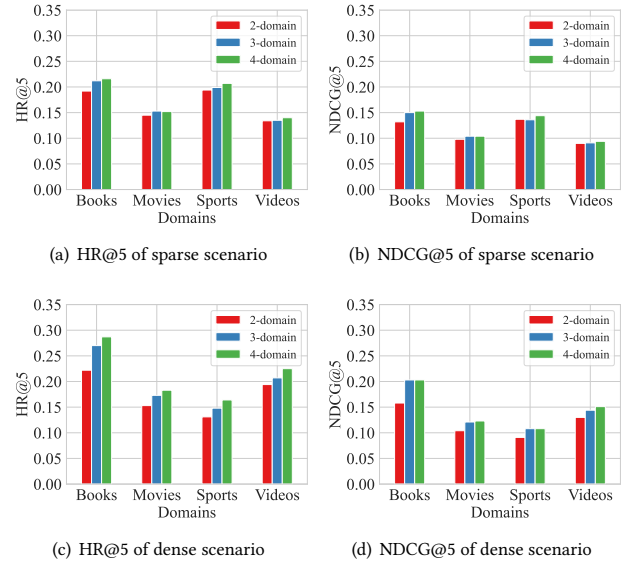


Figure 3: Effect of domain number in Amazon

at most several minutes for training, which is much faster than single-target and dual-target CDRs in most cases. This is because single-target (dual-target) CDRs require training individual models for all domains (all couples of domains) to achieve the best performance on all domains.

5.5.4 Impact of Domain Number. We last investigate the impact of the domain number N_d on the performance of MITrans. We measured MITrans under all combinations of 2, 3, and 4 domains for the sparse and target scenarios of Amazon dataset.

Figure 3 depicts the best performance of all combinations for each case. We see that increasing the number of domains improves the performance of MITrans in most cases. This observation indicates that MITrans effectively learns and leverages knowledge from multiple domains to improve its recommendation performance.

6 CONCLUSION

This work focused on a new CDR task, i.e., non-overlapped multi-target CDR, and proposed MITrans. In MITrans, we extract two types of representations for items from their text data and introduce two novel mutual information-based constraints, i.e., AFCLUB and AFDCE, to learn domain-specific and domain-shared preferences, respectively. Furthermore, we devised a preference fusing module to fuse domain-shared preferences learned in different domains and combine them with domain-specific preferences. We conducted extensive experiments on two real-world datasets. The results demonstrate that MITrans outperforms state-of-the-art non-overlapped CDR methods and achieves at least comparable performances with overlapped CDR methods. Moreover, MITrans empirically shows superiority in leveraging knowledge from multiple domains and alleviating negative transfer.

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REFERENCES

- [1] Daichi Amagata and Takahiro Hara. 2021. Reverse Maximum Inner Product Search: How to efficiently find users who would like to buy my item?. In *RecSys*. 273–281.
- [2] Daichi Amagata and Takahiro Hara. 2023. Reverse Maximum Inner Product Search: Formulation, Algorithms, and Analysis. *ACM Trans. on the Web* 17, 4 (2023), 1–23.
- [3] Kazuyoshi Aoyama, Daichi Amagata, Sumio Fujita, and Takahiro Hara. 2023. Simpler is Much Faster: Fair and Independent Inner Product Search. In *SIGIR*. 2379–2383.
- [4] Jiangxia Cao, Xin Cong, Jiawei Sheng, Tingwen Liu, and Bin Wang. 2022. Contrastive Cross-Domain Sequential Recommendation. In *CIKM*. 138–147.
- [5] Jiangxia Cao, Shaoshuai Li, Bowen Yu, Xiaobo Guo, Tingwen Liu, and Bin Wang. 2023. Towards Universal Cross-Domain Recommendation. In *WSDM*. 78–86.
- [6] Jiangxia Cao, Xixun Lin, Xin Cong, Jing Ya, Tingwen Liu, and Bin Wang. 2022. DisenCDR: Learning Disentangled Representations for Cross-Domain Recommendation. In *SIGIR*. 267–277.
- [7] Zhihong Chen, Rong Xiao, Chenliang Li, Gangfeng Ye, Haochuan Sun, and Hongbo Deng. 2020. ESAM: Discriminative Domain Adaptation with Non-Displayed Items to Improve Long-Tail Performance. In *SIGIR*. 579–588.
- [8] Pengyu Cheng, Weituo Hao, Shuyang Dai, Jiachang Liu, Zhe Gan, and Lawrence Carin. 2020. CLUB: A Contrastive Log-ratio Upper Bound of Mutual Information. In *ICML*, Vol. 119. 1779–1788.
- [9] Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, and Chong-Kwon Kim. 2022. Review-Based Domain Disentanglement without Duplicate Users or Contexts for Cross-Domain Recommendation. In *CIKM*. 293–303.
- [10] Qiang Cui, Tao Wei, Yafeng Zhang, and Qing Zhang. 2020. HeroGRAPH: A Heterogeneous Graph Framework for Multi-Target Cross-Domain Recommendation. In *RecSys*, Vol. 2715.
- [11] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT*. 4171–4186.
- [12] J. Du, Z. Ye, B. Guo, Z. Yu, and L. Yao. 2023. Distributional Domain-Invariant Preference Matching for Cross-Domain Recommendation. In *ICDM*. 81–90.
- [13] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor S. Lempitsky. 2016. Domain-Adversarial Training of Neural Networks. *JMLR* 17 (2016), 59:1–59:35.
- [14] Zhongxuan Han, Xiaolin Zheng, Chaochao Chen, Wenjie Cheng, and Yang Yao. 2023. Intra and Inter Domain HyperGraph Convolutional Network for Cross-Domain Recommendation. In *SIGIR*. 639–649.
- [15] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yong-Dong Zhang, and Meng Wang. 2020. LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. In *SIGIR*. 639–648.
- [16] Kohei Hirata, Daichi Amagata, Sumio Fujita, and Takahiro Hara. 2022. Solving Diversity-aware Maximum Inner Product Search Efficiently and Effectively. In *RecSys*. 198–207.
- [17] SeongKu Kang, Junyoung Hwang, Wonbin Kweon, and Hwanjo Yu. 2020. DE-RRD: A Knowledge Distillation Framework for Recommender System. In *CIKM*. 605–614.
- [18] Muhammad Murad Khan, Roliana Ibrahim, and Imran Ghani. 2017. Cross Domain Recommender Systems: A Systematic Literature Review. *ACM Comput. Surv.* 50, 3 (2017), 36:1–36:34.
- [19] Diederik P. Kingma and Max Welling. 2014. Auto-Encoding Variational Bayes. In *ICLR*.
- [20] Thomas N. Kipf and Max Welling. 2017. Semi-Supervised Classification with Graph Convolutional Networks. In *ICLR*.
- [21] Adit Krishnan, Mahashweta Das, Mangesh Bendre, Hao Yang, and Hari Sundaram. 2020. Transfer Learning via Contextual Invariants for One-to-Many Cross-Domain Recommendation. In *SIGIR*. 1081–1090.
- [22] Chenglin Li, Yuanzhen Xie, Chenyun Yu, Bo Hu, Zang Li, Guoqiang Shu, Xiaohu Qie, and Di Niu. 2023. One for All, All for One: Learning and Transferring User Embeddings for Cross-Domain Recommendation. In *WSDM*. 366–374.
- [23] Chenglin Li, Mingjun Zhao, Huanming Zhang, Chenyun Yu, Lei Cheng, Guoqiang Shu, Beibei Kong, and Di Niu. 2022. RecGURU: Adversarial Learning of Generalized User Representations for Cross-Domain Recommendation. In *WSDM*. 571–581.
- [24] Pan Li and Alexander Tuzhilin. 2020. DDTCDR: Deep Dual Transfer Cross Domain Recommendation. In *WSDM*. 331–339.
- [25] Siqing Li, Liuyi Yao, Shanlei Mu, Wayne Xin Zhao, Yaliang Li, Tonglei Guo, Bolin Ding, and Ji-Rong Wen. 2021. Debiasing Learning based Cross-domain Recommendation. In *KDD*. 3190–3199.
- [26] Zhi Li, Daichi Amagata, Takuya Maekawa, Kei Yonekawa, Mori Kurokawa, and Takahiro Hara. 2022. Trends-enhanced Attention & Memory Networks for E-commerce Recommendation. In *SIGIRCom workshop*.
- [27] Zhi Li, Daichi Amagata, Yihong Zhang, Takahiro Hara, Shuichiro Haruta, Kei Yonekawa, and Mori Kurokawa. 2022. Debiasing Graph Transfer Learning via Item Semantic Clustering for Cross-Domain Recommendations. In *IEEE BigData*. 762–769.
- [28] Zhi Li, Daichi Amagata, Yihong Zhang, Takahiro Hara, Shuichiro Haruta, Kei Yonekawa, and Mori Kurokawa. 2023. Semantic Relation Transfer for Non-overlapped Cross-domain Recommendations. In *PAKDD*. 271–283.
- [29] Zhi Li, Daichi Amagata, Yihong Zhang, Takuya Maekawa, Takahiro Hara, Kei Yonekawa, and Mori Kurokawa. 2022. HML4Rec: Hierarchical meta-learning for cold-start recommendation in flash sale e-commerce. *KBS* 255 (2022), 109674.
- [30] Jian Liu, Pengpeng Zhao, Fuzhen Zhuang, Yanchi Liu, Victor S. Sheng, Jiajie Xu, Xiaofang Zhou, and Hui Xiong. 2020. Exploiting Aesthetic Preference in Deep Cross Networks for Cross-domain Recommendation. In *WWW*. 2768–2774.
- [31] Weiming Liu, Xiaolin Zheng, Mengling Hu, and Chaochao Chen. 2022. Collaborative Filtering with Attribution Alignment for Review-based Non-overlapped Cross Domain Recommendation. In *WWW*. 1181–1190.
- [32] Weiming Liu, Xiaolin Zheng, Jiajie Su, Mengling Hu, Yanchao Tan, and Chaochao Chen. 2022. Exploiting Variational Domain-Invariant User Embedding for Partially Overlapped Cross Domain Recommendation. In *SIGIR*. 312–321.
- [33] Hayato Nakama, Daichi Amagata, and Takahiro Hara. 2021. Approximate Top-k Inner Product Join with a Proximity Graph. In *IEEE Big Data*. 4468–4471.
- [34] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation learning with contrastive predictive coding. *arXiv* (2018).
- [35] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars S. Thieme. 2009. BPR: Bayesian Personalized Ranking from Implicit Feedback. In *UAI*. 452–461.
- [36] Cheng Wang, Mathias Niepert, and Hui Li. 2020. RecSys-DAN: Discriminative Adversarial Networks for Cross-Domain Recommender Systems. *IEEE Trans. Neural Networks Learn. Syst.* 31, 8 (2020), 2731–2740.
- [37] Hanxin Wang, Daichi Amagata, Takuya Maekawa, Takahiro Hara, Hao Niu, Kei Yonekawa, and Mori Kurokawa. 2019. Preliminary Investigation of Alleviating User Cold-Start Problem in E-commerce with Deep Cross-Domain Recommender System. In *WWW*. 398–403.
- [38] Hanxin Wang, Daichi Amagata, Takuya Maekawa, Takahiro Hara, Hao Niu, Kei Yonekawa, and Mori Kurokawa. 2019. Preliminary Investigation of Alleviating User Cold-start Problem in E-commerce with Deep Cross-domain Recommender System. In *ECNLP workshop*. 398–403.
- [39] Junda Wu, Zhihui Xie, Tong Yu, Handong Zhao, Ruiyi Zhang, and Shuai Li. 2022. Dynamics-Aware Adaptation for Reinforcement Learning Based Cross-Domain Interactive Recommendation. In *SIGIR*. 290–300.
- [40] Kun Xu, Yuanzhen Xie, Liang Chen, and Zibin Zheng. 2021. Expanding Relationship for Cross Domain Recommendation. In *CIKM*. 2251–2260.
- [41] Chunfeng Yang, Huan Yan, Donghan Yu, Yong Li, and Dah Ming Chiu. 2017. Multi-site User Behavior Modeling and Its Application in Video Recommendation. In *SIGIR*. 175–184.
- [42] Wenhui Yu, Xiao Lin, Junfeng Ge, Wenwu Ou, and Zheng Qin. 2020. Semi-supervised Collaborative Filtering by Text-enhanced Domain Adaptation. In *KDD*. 2136–2144.
- [43] Feng Yuan, Lina Yao, and Boualem Benatallah. 2019. DAREC: Deep Domain Adaptation for Cross-Domain Recommendation via Transferring Rating Patterns. In *IJCAI*. 4227–4233.
- [44] Tianzi Zang, Yanmin Zhu, Haobing Liu, Ruohan Zhang, and Jiadi Yu. 2023. A Survey on Cross-domain Recommendation: Taxonomies, Methods, and Future Directions. *ACM Trans. Inf. Syst.* 41, 2 (2023), 42:1–42:39.
- [45] Shuai Zhang, Lina Yao, Aixin Sun, and Yi Tay. 2019. Deep Learning Based Recommender System: A Survey and New Perspectives. *ACM Comput. Surv.* 52, 1 (2019), 5:1–5:38.
- [46] Yinan Zhang, Yong Liu, Peng Han, Chunyan Miao, Lizhen Cui, Baoli Li, and Haihong Tang. 2020. Learning Personalized Itemset Mapping for Cross-Domain Recommendation. In *IJCAI*. 2561–2567.
- [47] Chuang Zhao, Hongke Zhao, Ming He, Jian Zhang, and Jianping Fan. 2023. Cross-domain recommendation via user interest alignment. In *WWW*. 887–896.
- [48] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. 2023. A Survey of Large Language Models. *arXiv preprint arXiv:2303.18223* (2023).
- [49] Yi Zhao, Chaozuo Li, Jiquan Peng, Xiaohan Fang, Feiran Huang, Senzhang Wang, Xing Xie, and Jibing Gong. 2023. Beyond the Overlapping Users: Cross-Domain Recommendation via Adaptive Anchor Link Learning. In *SIGIR*. 1488–1497.
- [50] Feng Zhu, Chaochao Chen, Yan Wang, Guanfang Liu, and Xiaolin Zheng. 2019. DTCDR: A framework for dual-target cross-domain recommendation. In *CIKM*. 1533–1542.
- [51] Feng Zhu, Yan Wang, Chaochao Chen, Guanfang Liu, and Xiaolin Zheng. 2020. A Graphical and Attentional Framework for Dual-Target Cross-Domain Recommendation. In *IJCAI*. 3001–3008.
- [52] Feng Zhu, Yan Wang, Chaochao Chen, Jun Zhou, Longfei Li, and Guanfang Liu. 2021. Cross-Domain Recommendation: Challenges, Progress, and Prospects. In *IJCAI*. 4721–4728.
- [53] Yongchun Zhu, Zhenwei Tang, Yudan Liu, Fuzhen Zhuang, Ruobing Xie, Xu Zhang, Leyu Lin, and Qing He. 2022. Personalized Transfer of User Preferences for Cross-domain Recommendation. In *WSDM*. 1507–1515.